Project Title

Enterprise AI SQL Analytics Platform using Agentic AI

Project Overview

The project aims to build an AI-powered database analytics platform that allows users to interact with organizational databases using natural language instead of writing SQL queries manually.

Users can ask questions such as:

- Show the top 5 customers by revenue.
- Which products generated the highest sales in the last 6 months?
- Show monthly sales trends.

The system automatically understands the request, generates SQL queries, retrieves data, analyzes results, and presents insights through charts and summaries.

The objective is to make database analytics accessible to both technical and non-technical users.

Problem Statement

Most organizations depend on database administrators or developers to write SQL queries and generate reports.

This project removes that dependency by allowing users to communicate with databases using natural language and receive meaningful insights instantly.

Core Idea

Traditional systems:

User → SQL Expert → Database → Report

Proposed System:

User → AI Agent System → Database → Insights + Visualizations

The system acts as an intelligent data analyst capable of understanding business questions and generating meaningful reports.

Agentic AI Workflow

User Question ↓ Schema Agent ↓ SQL Generator Agent ↓ SQL Validation Agent ↓ Query Execution ↓ Insight Agent ↓ Visualization Agent ↓ Final Answer + Charts

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Main Features

1. Natural Language to SQL

Users ask questions in plain English.

Example:

"Show customers who spent more than ₹10,000 this year."

The system automatically generates and executes SQL queries.

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2. Automatic Schema Understanding

The AI automatically analyzes:

- Tables
- Columns
- Relationships

Users do not need to know database structure.

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3. SQL Validation

Generated queries are validated before execution.

The system checks:

- Incorrect table names
- Invalid columns
- Syntax issues

This improves reliability and reduces errors.

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4. Query Explanation

Before execution, the system explains the generated query in simple language.

Example:

"This query retrieves customers whose yearly spending exceeds ₹10,000."

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5. Result Interpretation

Instead of displaying only raw tables, the AI explains the business meaning behind the results.

Example:

"Product A contributed 32% of total sales and is currently the highest revenue-generating product."

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6. Automatic Data Visualization

The system automatically generates:

- Bar Charts
- Pie Charts
- Line Charts
- Trend Analysis Graphs

based on the returned data.

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7. Query History

Users can view:

- Previous questions
- Generated SQL
- Execution timestamps
- Generated insights

User Roles

Admin

Responsibilities:

- Configure database connection
- Create employee accounts
- Assign predefined roles
- Manage users
- Access SQL Workspace
- Execute CRUD operations manually
- Commit transactions
- Rollback transactions

Admin SQL Permissions

Allowed:

- SELECT
- INSERT
- UPDATE
- DELETE
- COMMIT
- ROLLBACK

The SQL Workspace is intended for technical administrators who already have authority over the connected database.

Employee

Responsibilities:

- Login
- Ask analytical questions
- View charts
- View query history

Employee SQL Permissions

Allowed:

- SELECT only

Employees cannot modify organizational data.

Security & Privacy

Privacy is a major design goal.

Key Principles

- Organizational data is never stored by the platform.
- Companies connect their own databases.
- Data remains within the organization.
- Database credentials are stored securely.
- Organizations are isolated from one another.

Data Safety

AI-generated queries are restricted to read-only operations.

Allowed:

- SELECT

Blocked:

- INSERT
- UPDATE
- DELETE
- DROP

Only Admin users can manually execute modification queries through the SQL Workspace.

This prevents accidental damage caused by incorrect AI-generated queries.

Technology Stack

Frontend

- React
- Tailwind CSS

Backend

- FastAPI

AI Layer

- LangChain
- LangGraph

LLM

- Ollama
- Qwen 3 8B

Database

- PostgreSQL

Charts

- Recharts

Authentication

- JWT

System Architecture

React Frontend ↓ FastAPI Backend ↓ LangGraph Workflow

Schema Agent

SQL Generator Agent

SQL Validation Agent

Insight Agent

Visualization Agent

↓ PostgreSQL Database ↓ Charts + Business Insights

Project Scope (Optimized for 1.5 Months)

Must Have Features

- User Authentication
- Admin & Employee Roles
- Database Connection Setup
- Natural Language to SQL
- Schema Understanding
- SQL Validation
- Query Execution
- Result Interpretation
- Data Visualization
- Query History
- Admin SQL Workspace
- Commit & Rollback Support

Nice to Have Features

- PDF Report Export
- Dashboard Generation
- Agent Workflow Visualization

Out of Scope

- Voice Commands
- Multi-Database Support
- Real-Time Monitoring
- Complex Custom Permission Systems

Expected Outcome

A fully functional Enterprise AI SQL Analytics Platform that enables organizations to interact with databases using natural language while maintaining privacy, security, and ease of use.

The project demonstrates:

- Agentic AI
- LangChain
- LangGraph

- Natural Language Processing
- Database Analytics
- Data Visualization
- Full Stack Development
- Role-Based Access Control

while remaining realistic to complete within approximately 1.5 months alongside academic coursework.